

MTL1002
Linear Algebra and Differential Equations
Department of Mathematics
Indian Institute of Technology Delhi

Course Content:

Linear Algebra:

Systems of linear (homogeneous and non-homogeneous) equations, matrices and Gauss elimination, elementary row operations, row reduced echelon form, row space, column space, null space, and rank of a matrix. Vector spaces over $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, subspaces, linear independence, linear span of a set of vectors, basis and dimension of a vector space, sum and direct sum. Introduction to inner product and norm (definitions and examples), linear transformation, rank-nullity theorem and its applications, matrix representation of a linear transformation, change of basis, and similarity. Eigenvalues and eigenvectors, characteristic polynomial, Cayley-Hamilton theorem (without proof) and applications.

Differential Equations:

Review of first-order differential equations, existence and uniqueness of solutions of an initial value problem, Picard's theorem. Linear independence of solutions of linear differential equations, dimensionality of the space of solutions, and Wronskian. Linear ODE with constant coefficients of second and higher order, Cauchy-Euler equations, method of undetermined coefficients and method of variation of parameters. System of linear differential equations with constant coefficients, fundamental matrix, matrix methods. Taylor series method, Laplace transform and its application to solve ODE.

References:

- Notes For Linear Algebra:: <http://web.iitd.ac.in/~ritumoni/courses.html>
- Notes for Differential Equations: <https://web.iitd.ac.in/~priyadarshi/>
- Advanced Engineering Mathematics, E. Kreyszig (John Wiley).
- Introduction to Linear Algebra by Gilbert Strang (South Asian Edition).

Exams Weightage:

Exam	Weightage
Quiz 1	15%
Quiz 2	15%
Minor	30%
Major	40%

Tentative Exam Schedule:

Exam	Date and time	Tentative syllabus
Quiz 1	February 03, 2026 (at 6:00pm)	Will be announced
Minor	February 20-25, 2026 (Refer to Institute time table)	All Linear Algebra
Quiz 2	April 07, 2026 (6:00pm)	Will be announced
Major	April 29-May 05, 2026 (Refer to Institute time table)	Entire syllabus

Attendance: As per institute rule.

Grading Policy: The passing grade (D grade) will at 30%.
An A grade will be at 80% or higher (will be decided later).
All other grades will be decided later.

Re-quiz: If you have missed the one or both the quizzes due to medical reasons there will be one re-quiz (only ONE). Submit your medical certificate (IITD Hospital) to your class teacher / TA within two working days after the quiz.

Re-minor: If you have missed your minor exam due to medical reasons there will be a re-minor exam sometime before the major exam. Submit your medical certificate (IITD Hospital) to your class teacher / TA within two working days of the minor exam.